

Nicholas Lee

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Skills

CAD & CAE: SolidWorks, COMSOL Multiphysics, ANSYS, AutoCAD, Onshape, FEA, CFD, Surface Modeling, DFM/DFA

Manufacturing: 3D Printing (FDM), Sheet Metal Design, Milling, Wet Layup, Root Cause Analysis, BOM Development

Programming: Python (Pandas, NumPy, Matplotlib), MATLAB/Simulink, C++ , Git

Experience

Research Intern, Harvard University, Aizenberg Lab – Cambridge, MA May 2026 – Present

- Ran CFD simulations in COMSOL Multiphysics across multiple nasal-cavity geometries to quantify how anatomy affects airflow, mass transport, and odor detection for a universal adaptive electronic nose
- Built Python scripts (Pandas, NumPy) to process large CFD datasets, extracting airflow and mass-transfer metrics for reproducible comparison across geometries
- Validated novel low-cost methane sensors in a field study, benchmarking their output against industry-standard equipment to confirm real-world accuracy

Mechanical Designer, Midnight Sun Solar Car Team – Waterloo, ON Sept 2025 – Present

- Reduced ballast-box weight by 20% while maintaining structural integrity by designing an aluminum sheet-metal enclosure in SolidWorks using DFM/A principles and static FEA
- Improved wheel-cover assembly accuracy and reduced installation errors by modeling cutting stencils referenced to vehicle datum features
- Cut camera-mount lead time from weeks to minutes by designing and prototyping a press-fit, 3D-printable mount system

Mechanical Team Member, Waterloo Automation Collective – Waterloo, ON Jan 2026 – Present

- Accelerated design iteration and centralized CAD management by developing parametrically driven clamping linkages in Onshape
- Drove early-stage design and prototyping of the clamping subsystem as a founding member, building a desktop injection molder

Undergraduate Research Assistant, MSAM Lab – Waterloo, ON Jan 2026 – Apr 2026

- Reduced gear design cycle time by developing parametric SolidWorks CAD models that enabled rapid, repeatable design changes
- Ensured reliable operation of educational gearbox assemblies by iteratively designing SolidWorks components optimized for FDM 3D printing
- Enhanced hands-on educational demonstrations by integrating DC motors into gearbox systems to enable automated actuation

Engineering Intern, Refresco Beverages – Mississauga, ON Jan 2025 – Aug 2025

- Extended pneumatic piston assembly service life by 200% by reverse-engineering components in SolidWorks to mitigate sugar buildup
- Reduced unplanned downtime and improved plant efficiency by developing technical documentation, assembly drawings, and BOMs integrated into the plant CMMS

Education

University of Waterloo – BASc in Mechatronics Engineering

Sept 2023 – May 2028

GPA: 3.85 – Dean's Honours List

Relevant Coursework: Mechanics of Deformable Solids, Thermodynamics and Heat Transfer, Design and Dynamics of Machines